



Medicaid In-Patient (IP) Model Refresh

Elle Palmer



Problem

- Current inpatient (IP) model was built in 2022 with the intent of being a temporary model
- New data and data processing methods have become available since the IP model was built (ex: claims transformer embeddings [TE] model) that are not currently leveraged
- Data platform migration in 2023/2024 makes model optimization, monitoring, and management easier

Approach

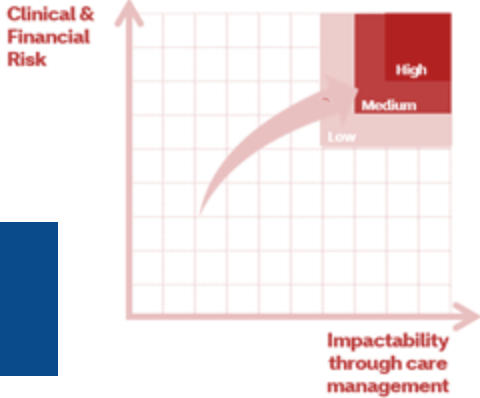
- Leverage existing model building pipelines to test different methodologies and refresh the IP model
- Investigate easy-to-implement explainability methods for TEs
- Collaborate with data engineering (DE) to deploy the model with monitoring services

Outcome

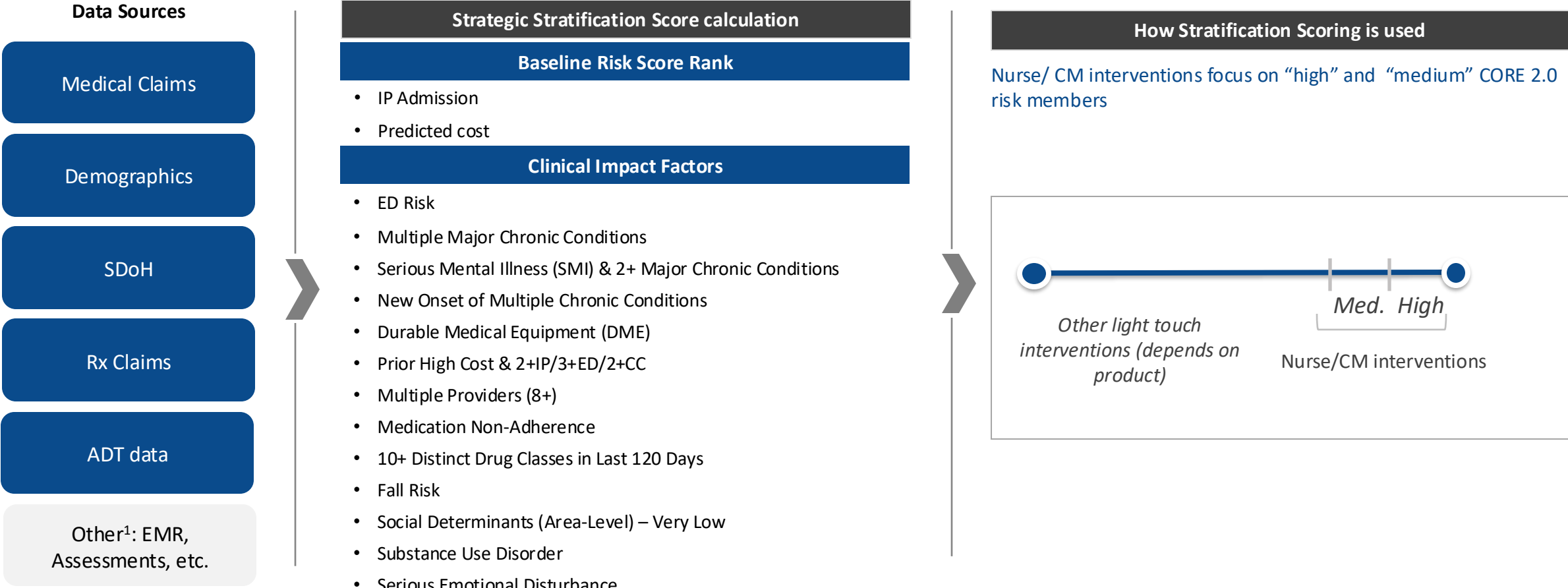
- Reusable model training pipelines (catboost, xgboost) are nearly complete for shared use
- New IP model has a performance boost of ~1.3x the 2022 model in top 1%
- 10-20% of TEs have some explainability by leveraging correlation pattern clustering
- DE intern built out a reusable model monitoring pipeline that will be deployed with the Medicaid IP model from day 1

Background/model development

CORE 2.0 Score



CORE 2.0 leverages **baseline predictive risk scores** and **clinical impact factors** to identify and prioritize members who have **high clinical and financial risk** and are **impactable through care management**



¹. Future state
4 ©2021 CVS Health and/or one of its affiliates. Confidential and proprietary.

Embeddings explainability overview

Feature Correlations

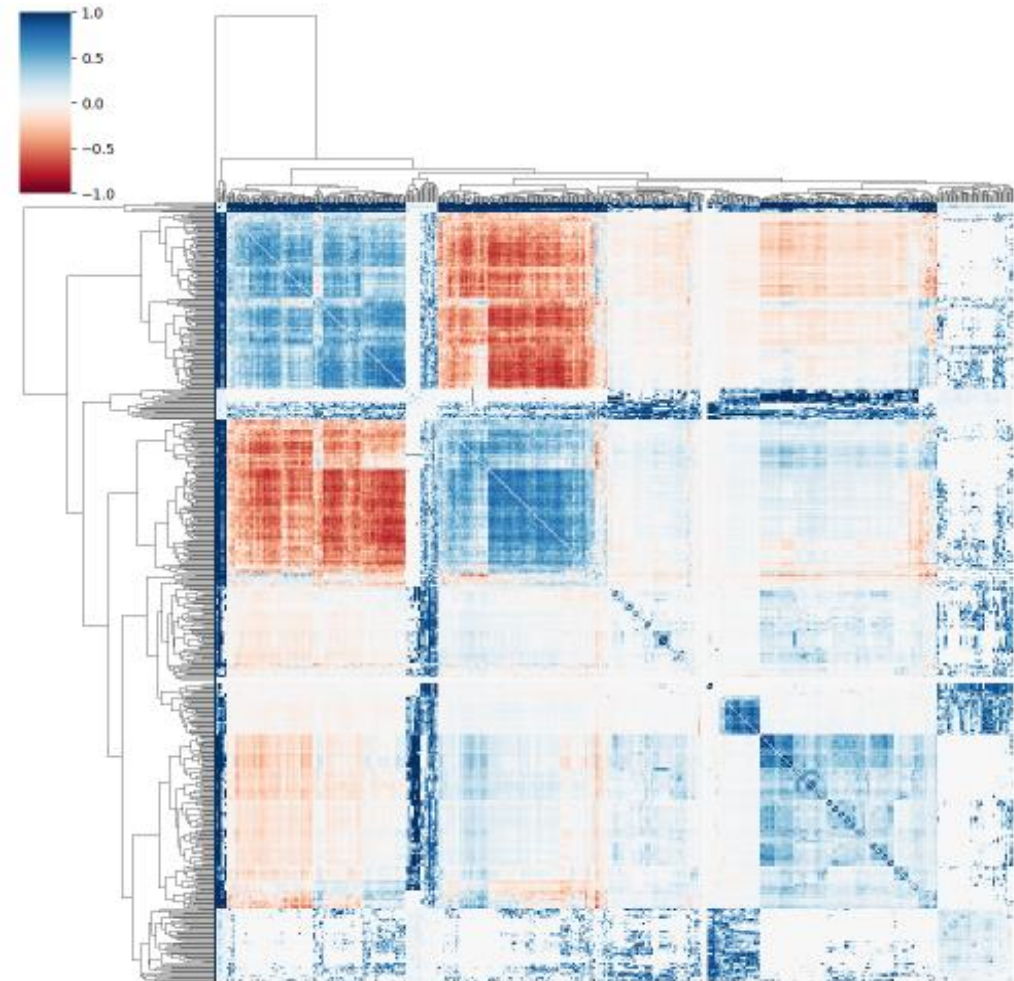
- **561 features included:**

- Chronic conditions: (n = 52)
- Demographics (n = 7)
- Social determinants: (n = 29)
- Claims and Utilization: (n = 138 Medical, n = 79 Rx)
- Claims transformer embedding features: (n = 256)

- **Correlation/Association Methods**

- Pearson's Correlation: Continuous to Continuous
- Anova: Continuous to Categorical
- Point biserial: Continuous to Binary
- Cramer's V: Categorical to Categorical

Cross-feature correlations



TE Interpretability: Correlation clustering allows for interpretation of 10-20% of claims TE features

- Many of the embeddings cluster with other embeddings or represent a unique signal we need to do more work to understand
- Embedding 224 is the 4th most important in Catboost but **does not cluster with any traditional features or other embeddings**
- Some notable embeddings with interpretations that were higher on the feature importance list for the best Catboost model
 - Embedding 58 is correlated with the SDoH indicator for **health infrastructure**
 - Embedding 94 is correlated with CMS preventative care measure for **pelvic exam in past 12 months and low/medium severity ED use in 13-24 months pre-period**

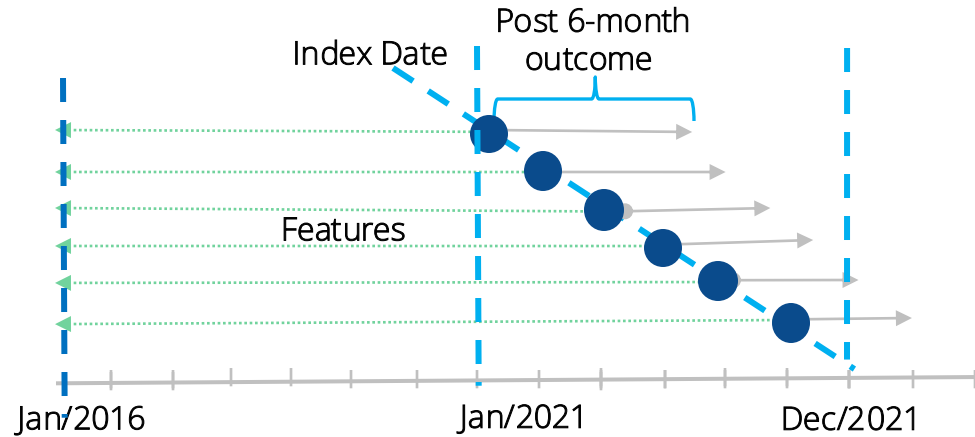
329	79	emb24
363	79	emb58
346	79	emb41
347	79	emb42
395	79	emb90
382	79	emb77
340	79	emb35
274	79	health_infra

511	22	emb206
302	22	cms_pelvic
318	22	emb13
337	22	emb32
399	22	emb94
505	22	emb200
493	22	emb188
29	22	low_med_sev_ed_flag_yr2

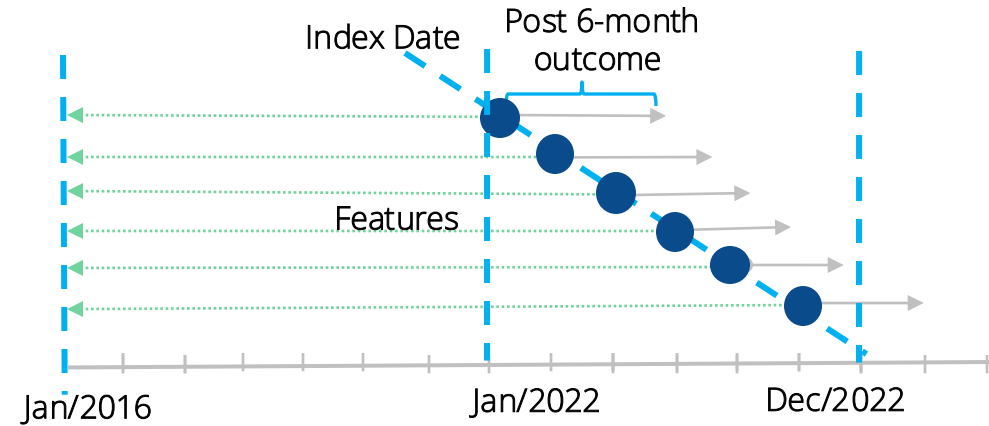
Model Development

Clinical oversight, model setup and out of time (OOT) validation

Model Training



Out of Time Validation



- **Clinical oversight on IP definition - meeting with clinical partners**
 - Inclusion criteria: Acute medical inpatient (COE 10200, 10700, 10800)
 - Exclusion criteria: IP medical case due to pregnancy / birth / transplant
- **Model setup**
 - Modeling sample: active members in Oct 2022-Sep 2023, select a random membership month for each member to train on full year
 - Outcome variable: Post 6 months acute IP event from index date
 - Split the sample into Training/Validation/Test per 80%:10%:10%
- **OOT validation**
 - Nov 2023 membership

Data sources and features

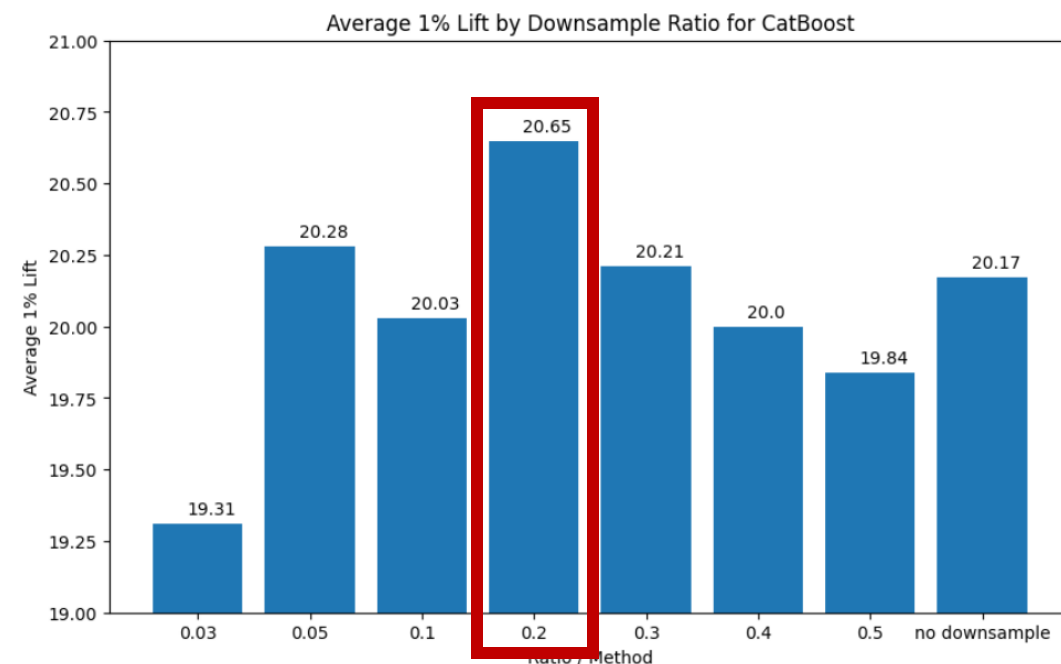
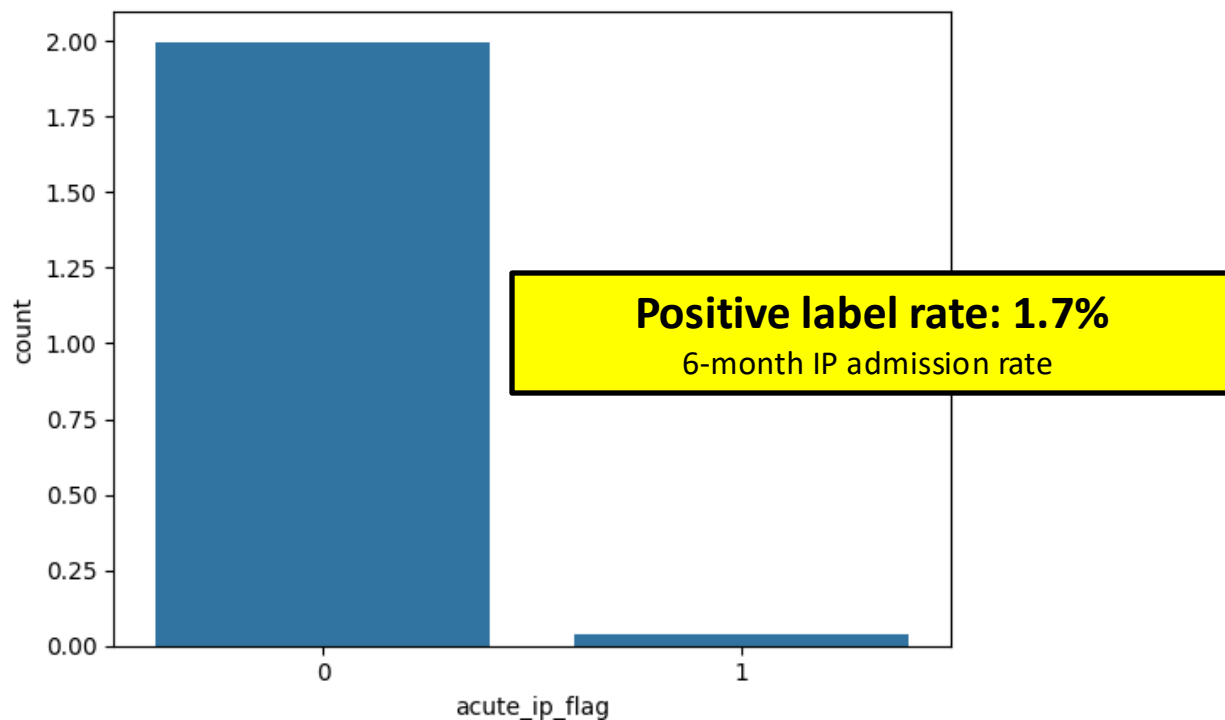
Feature Space			
Features	By	Legacy	Enhanced
Diagnosis Codes	Transformer embedding	N	Y
Procedure codes		N	Y
Provider specialty		N	Y
Revenue code		N	Y
Place of service		N	Y
Chronic conditions	Traditional feature engineered	Y	Y
IP utilization 0-24 months prior		Y	Y
ED utilization 0-24 months prior		Y	Y
OP utilization 0-24 months prior		Y	Y
Rx utilization 0-24 months prior		Y	Y
Preventative care		N	Y
Age, gender, membership length		Y	Y
Enterprise SDOH			
-- CSDI indices		N	Y
-- ACS		Y	Y
-- BFRSS		Y	N

Data was pulled from BQ and modeling completed using Vertex AI. Features focus on members' clinical conditions, healthcare journey, demographic, and geographic characteristics.

- **Transformer Embedding**: 256 features, up to 24-month history by index date
- **Traditionally engineered features**:
 - **Other clinical**: PPM conditions, Rx, utilization counts aggregated at 0-12- and 13-24-month prior level
 - **Demographic**: age, gender, membership length
 - **Geographic**: enterprise SDoH at census tract level of prior year

Preprocessing

- Several steps were taken to ensure data could best be processed by the model
 - Filling in null or empty values
 - Determining and encoding categorical values using one-hot
 - Downsampling due to data imbalance**

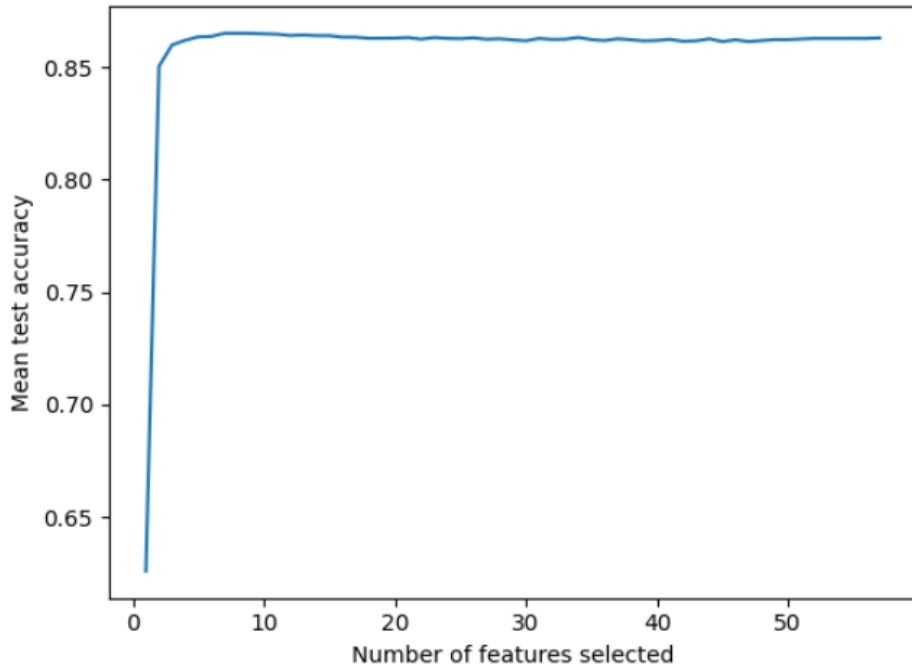


- Ran base model on a continuum of downsampled outcomes
- Pulled the average 1% lift for each ratio across random seeds
- Selected optimal ratio to use for the rest of the model pipeline

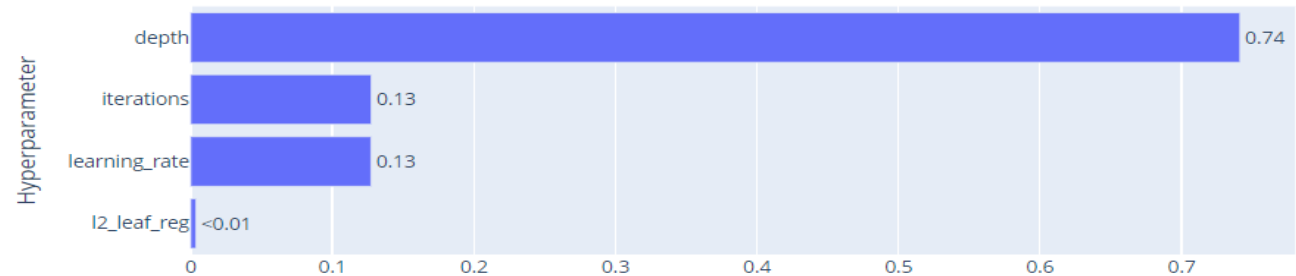
Model Optimization

- Ran Recursive Feature Elimination (RFE) on all features (embedding and traditional)
 - Catboost kept 499 features while XGBoost only kept 78 features
- Used Optuna to select optimal hyperparameters
- Depth and learning rate were the most important hyperparameters for CatBoost and XGBoost, respectively

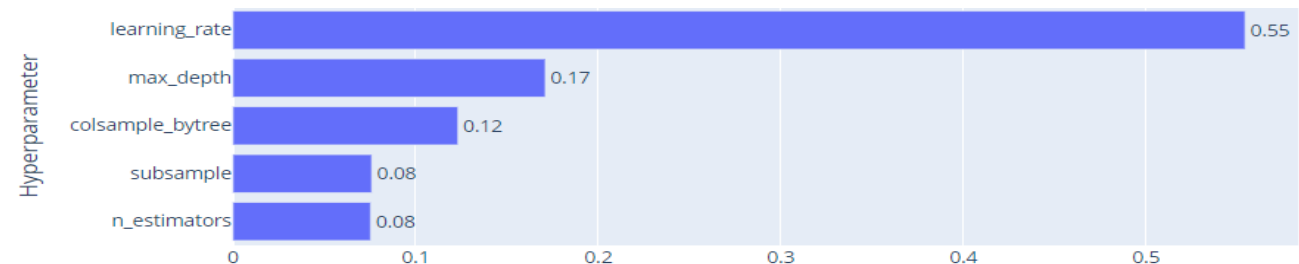
Recursive Feature Elimination
with correlated features



XGBoost Feature Elimination Curve



CatBoost Hyperparameter Importances



XGBoost Hyperparameter Importances

Results

Model Performance

- Initial results show the model performing very well, performing ~1.3x better than the current model in production in the top 1%
- OOT validation performance is slightly above test-holdout, similar to what Commercial saw
- Top features are similar between models, suggesting feature creation is more impactful than modeling method; TEs are the top features other than age (XGBoost and CatBoost)

Refreshed Model vs. Product Spec	1%				10%		
	AUC	Precision	Recall	Lift	Precision	Recall	Lift
XGBoost - Claim TEs and Engineered Features	0.87	36.7%	22.7%	22.7	10.5%	65.1%	6.5
XGBoost – Claim TEs only	0.85	33.1%	20.5%	20.5	9.9%	61.5%	6.2
CatBoost - Claim TEs and Engineered Features*	0.87	33.5%	19.5%	19.6	10.8%	60.4%	6.3
Neural Net - Claim TEs and Engineered Features*	0.85	34.1%	19.9%	19.9	10.5%	61.71%	6.2
Production (<i>same OOT set as XGBoost models</i>)	0.84	25.6%	17.6%	17.6	8.5%	58.5%	5.9

Medicaid Enhanced Inpatient Model

Background and Objective

- The current Inpatient model was implemented as a temporary improvement over the previous model developed by Medicaid Informatics
- New modeling methods, like Transformer Embeddings, incorporate longitudinal clinical information resulting in better predictive accuracy
- The inpatient model is one of the main factors in the CORE 2.0 risk score, which stratifies the entire population monthly and is used to identify high-risk members for care management

Progress & Next Steps

- Model will be completed by end of September and in production in December
- Initial results show the model performing very well, performing ~1.3x better than the current model in production in the top 1%

Utilization notes

- This model will only be used in the current process so there are no changes in workflows or IT

Refreshed vs. Product Spec – out of time validation				1%		10%	
	AUC	PPV	Sensitivity	Lift	PPV	Sensitivity	Lift
XGBoost - Claim TEs and Engineered Features	0.87	36.7%	22.7%	22.7	10.5%	65.1%	6.5
XGBoost – Claim TEs only	0.85	33.1%	20.5%	20.5	9.9%	61.5%	6.2
Production (same test set as new models)	0.84	25.6%	17.6%	17.6	8.5%	58.5%	5.9

Model performance is generalized across COAs

- A subgroup analysis allows us to verify we are not biasing our predictions towards or away from any aid group
- <1% of kids have an IP event in the following 6-months, leading to the lower PPV in that group

Refreshed vs. Product Spec – out of time validation	1%			10%		
	PPV	Sensitivity	Lift	PPV	Sensitivity	Lift
ABD (CORE 2.0 cohort)	34.7%	22.5%	22.5	10.1%	65.4%	6.5
Medicaid Adult (CORE 2.0 cohort)	42.3%	19.2%	19.2	13.6%	61.6%	6.2
Medicaid Kid (CORE 2.0 cohort)	16.3%	26.6%	26.6	3.6%	59.4%	5.9
Behavioral Health	36.7%	22%	21.9	10.6%	63.5%	6.3
Duals	31.1%	22.2%	22.1	8.3%	59.3%	5.9
Fosters	37.7%	23.3%	23.3	10.9%	67.5%	6.8
LTSS	39.9%	23.7%	23.7	11.2%	66.6%	6.7
Overall	36.7%	22.7%	22.7	10.5%	65.1%	6.5

Model performance across Membership length is acceptable

- Newer members are harder to predict because we have less data
- Even with this limitation, we do well capturing future IP events in new members

Refreshed vs. Product Spec – out of time validation	1%			10%		
	PPV	Sensitivity	Lift	PPV	Sensitivity	Lift
New member 0-3 month history	29.3%	16.9%	16.8	9%	51.7%	5.2
New member 4-8 month history	29.2%	17.3%	17.3	9.4%	55.8%	5.6
Established member 9+ month history	37.1%	23.1%	23.1	10.6%	66.1%	6.6
Overall	36.7%	22.7%	22.7	10.5%	65.1%	6.5

For the 6-month IP model, among the 78 selected features, 87% are clinical related

Feature Type	# Features	% Features
Rx	9	12%
Clinical	36	46%
Demographic	9	12%
Emddding	23	29%
SDoH	1	1%
Grand Total	78	100%

SDoH and Demographics	
Feature	Importance
agenbr	33,302
water_quality	22,948
tenure_yr1	16,375
gender	4,165
coa_population_group_ABD	3,646
coa_population_group_BH	3,511
coa_population_group_TANF/CHIP	3,414
coa_population_group_Expansion	3,311
coa_population_group_Dual Elig	3,064
coa_population_group_LTSS	2,166

Top 20 clinical features		
Rank	Feature	Importance
26	sum_op_visits_yr1	20,103
27	coe_other_clm_yr1	16,711
28	emis_ed_clm_yr1	16,528
30	coe_eval_clm_yr1	16,184
31	retail_fills_yr1	15,929
32	emis_spec_clm_yr1	15,166
33	emis_misc_clm_yr1	14,585
34	generic_fills_yr1	14,225
35	rx_claim_cnt_yr2	14,130
36	major_chronic_cnt	13,673
37	formulary_fills_yr1	13,303
38	coe_mrx_clm_yr1	12,565
39	coe_radio_clm_yr1	12,493
40	maint_drug_fills_yr2	12,465
41	sum_spec_yr1	12,169
42	emis_ambul_clm_yr2	11,828
43	emis_ip_clm_yr1	10,986
44	emis_hh_clm_yr1	10,225
45	anticonvulsant_days_supply_yr1	9,161
46	sum_ed_visits_yr1	8,250

Model monitoring plan

Use-case POC: 3-month in-patient model

Commercial inpatient model is used to predict member admissions within a specified timeframe into the future

Utilizes **transformer embeddings** trained to capture “**member journey**” along with clinical features

Deployed and running in production since **May 2024**

Performed observability evaluations on a **20% sample** with **3 million** rows of data with **256 features** (clinical + embeddings)

Key observability findings:

For the **top 1%** of cases (individuals with highest in-patient risk):

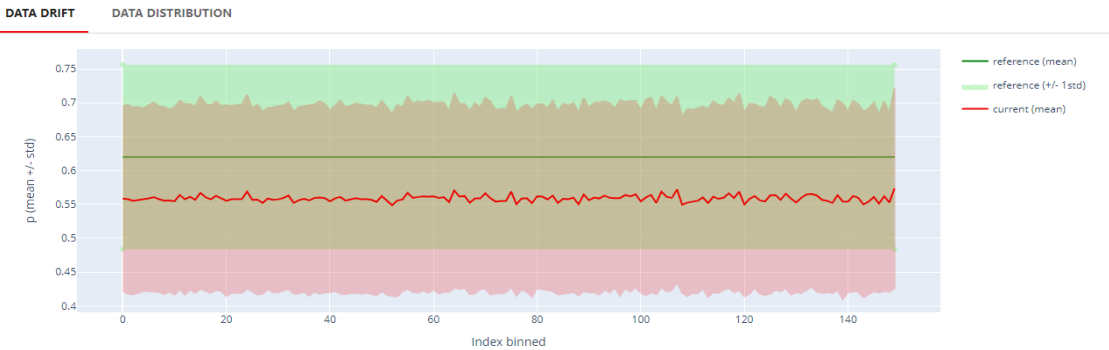
- **5%** of the features were detected to be **drifted** (PSI score > 0.1)
- **Prediction drift detected** in outcome prob. distribution (PSI score = 1.178)
- **No drift** detected in the 142-dimension **embeddings space** (cosine distance < 0.5)

Drift is detected for 4.669% of columns (12 out of 257).

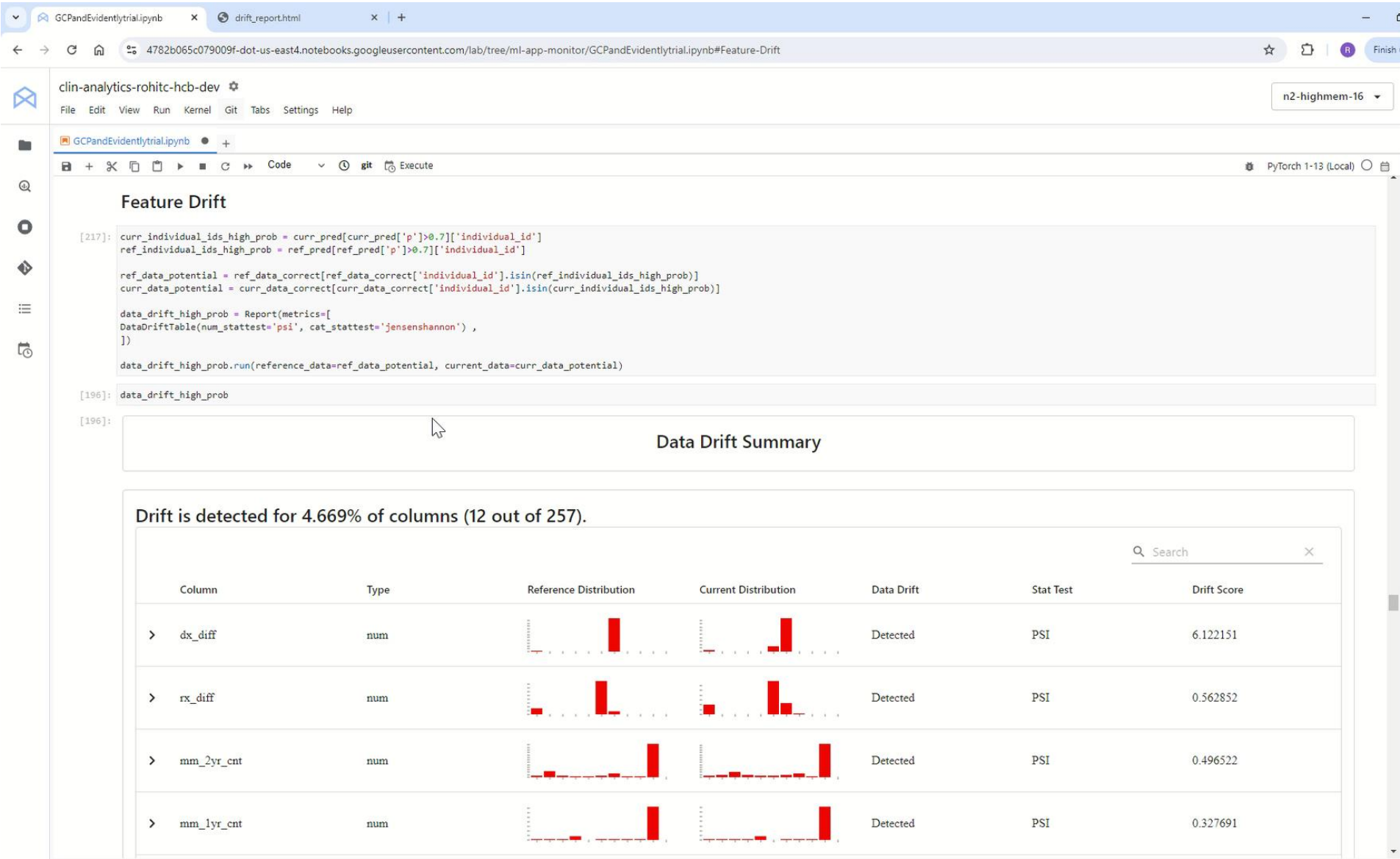
Column	Type	Reference Distribution	Current Distribution	Data Drift	Stat Test	Drift Score
> dx_diff	num			Detected	PSI	6.122151
> rx_diff	num			Detected	PSI	0.562852
> mm_2yr_cnt	num			Detected	PSI	0.496522
> mm_1yr_cnt	num			Detected	PSI	0.327691
> emb203	num			Detected	PSI	0.148192

Drift in column 'p'

Data drift detected. Drift detection method: PSI. Drift score: 1.178



Demo



Next steps

- 1. Code review**
- 2. PQI validation / DRG impactability assessment**
 - i. Verify top picks are truly impactable*
- 3. Bias check**
 - i. Race/ethnicity and gender*
- 4. CORE integration SAI valuation check**
- 5. DE handoff**
- 6. Product spec**

Questions?

Understanding Key Factors in Prediction Drift

Subset of individuals with the largest difference outcome probabilities (**low risk** -> **high risk** and vice versa)

- 65% of features drifted including **majority of transformer embeddings**
- Indicates that the embeddings effectively reflect patient journey

Top clinical features for increased risk

Chronic conditions

Unique procedure code counts – 3 months

Unique diagnostic codes – 3 months

Specialist visit count- 1 year

Revenue code counts – 3 months

ER claims counts – 3 months



Top clinical features for decreased risk

Unique procedure code counts – 3 months

Revenue code counts – 6 months

Member claims count – 3 months

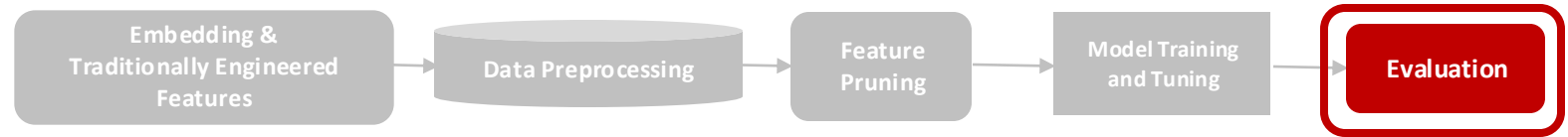
Lab visit counts – 3 months

Histrionic personality disorder (HPD) counts

Changes in prescriptions (pharmacy_gpi4) – 1 year



Feature Importances



- The 10 most important features for XGBoost and CatBoost show us that age is the strongest feature in both models
- Embeddings are more important in XGBoost than in CatBoost
- Catboost kept 499 features while XGBoost only kept 59

XGBoost Feature Rank	Feature Name
1	agenbr
2	Emb58
3	Emb212
4	Emb154
5	Emb238
6	Emb138
7	Emb31
8	Emb47
9	Emb7
10	Emb233

CatBoost Feature Rank	Feature Name
1	agenbr
2	coa_population_group
3	tenure_yr1
4	emb224
5	coe_eval_clm_yr1
6	major_chronic_cnt
7	emis_spec_clm_yr1
8	primarylanguage_desc
9	water_quality
10	emb81